

Quiz

1. Compared to analog systems, digital systems
 - a. are less prone to noise
 - b. can represent an infinite number of values
 - c. can handle much higher power
 - d. all of the above

Quiz.

2. The number of values that can be assigned to a bit are
- a. one
 - b. two
 - c. three
 - d. ten

Quiz

3. The time measurement between the 50% point on the leading edge of a pulse to the 50% point on the trailing edge of the pulse is called the

- a. rise time
- b. fall time
- c. period
- d. pulse width

Quiz

4. The time measurement between the 90% point on the trailing edge of a pulse to the 10% point on the trailing edge of the pulse is called the

- a. rise time
- b. fall time
- c. period
- d. pulse width

Quiz.

5. The reciprocal of the frequency of a clock signal is the
- a. rise time
 - b. fall time
 - c. period
 - d. pulse width

Quiz

6. If the period of a clock signal is 500 ps, the frequency is
- a. 20 MHz
 - b. 200 MHz
 - c. 2 GHz
 - d. 20 GHz

Quiz

7. AND, OR, and NOT gates can be used to form
- a. storage devices
 - b. comparators
 - c. data selectors
 - d. all of the above

Quiz

8. A shift register is an example of a
- a. storage device
 - b. comparator
 - c. data selector
 - d. counter

Quiz

9. A device that is used to switch one of several input lines to a single output line is called a

- a. comparator
- b. decoder
- c. counter
- d. multiplexer

Quiz.

10. For most digital work, an oscilloscope should be coupled to the signal using

- a. ac coupling
- b. dc coupling
- c. GND coupling
- d. none of the above

Quiz

1. For the binary number 1000, the weight of the column with the 1 is

- a. 4
- b. 6
- c. 8
- d. 10

Quiz

2. The 2's complement of 1000 is

- a. 0111
- b. 1000
- c. 1001
- d. 1010

Quiz

3. The fractional binary number 0.11 has a decimal value of

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{4}$

d. none of the above

Quiz

4. The hexadecimal number 2C has a decimal equivalent value of

- a. 14
- b. 44
- c. 64
- d. none of the above

Quiz

5. Assume that a floating point number is represented in binary. If the sign bit is 1, the

- a. number is negative
- b. number is positive
- c. exponent is negative
- d. exponent is positive

Quiz

6. When two positive signed numbers are added, the result may be larger than the size of the original numbers, creating overflow. This condition is indicated by

- a. a change in the sign bit
- b. a carry out of the sign position
- c. a zero result
- d. smoke

Quiz.

7. The number 1010 in BCD is
- a. equal to decimal eight
 - b. equal to decimal ten
 - c. equal to decimal twelve
 - d. invalid

Quiz.

8. An example of an unweighted code is
- a. binary
 - b. decimal
 - c. BCD
 - d. Gray code

Quiz.

9. An example of an alphanumeric code is
- a. hexadecimal
 - b. ASCII
 - c. BCD
 - d. CRC

Quiz

10. An example of an error detection method for transmitted data is the

- a. parity check
- b. CRC
- c. both of the above
- d. none of the above

Quiz

1. The truth table for a 2-input AND gate is

a.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

b.

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

c.

Inputs		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

d.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Quiz

2. The truth table for a 2-input NOR gate is

a.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

b.

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

c.

Inputs		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

d.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Quiz

3. The truth table for a 2-input XOR gate is

a.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

b.

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

c.

Inputs		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

d.

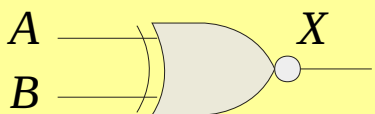
Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Quiz

4. The symbol $\begin{array}{c} A \text{ ---} \\ B \text{ ---} \end{array} \boxed{\geq 1} \text{ ---} X$ is for a(n)

- a. OR gate
- b. AND gate
- c. NOR gate
- d. XOR gate

Quiz.

5. The symbol  is for a(n)

- a. OR gate
- b. AND gate
- c. NOR gate
- d. XOR gate

Quiz

6. A logic gate that produces a HIGH output only when all of its inputs are HIGH is a(n)

- a. OR gate
- b. AND gate
- c. NOR gate
- d. NAND gate

Quiz

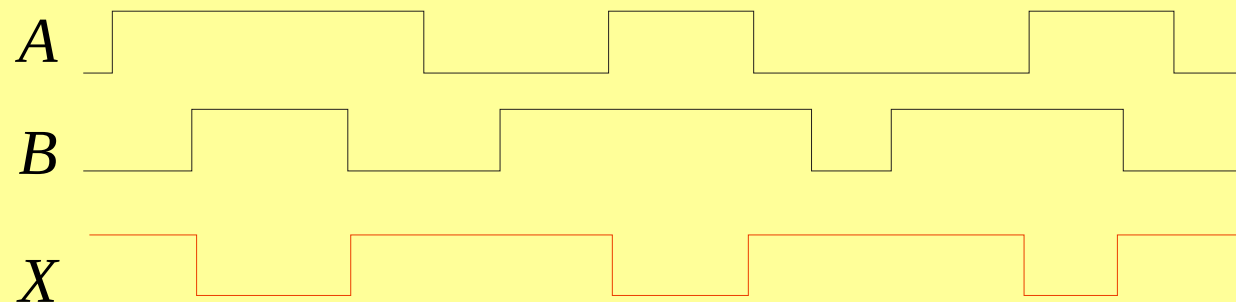
7. The expression $X = A \oplus B$ means

- a. $A \text{ OR } B$
- b. $A \text{ AND } B$
- c. $A \text{ XOR } B$
- d. $A \text{ XNOR } B$

Quiz

8. A 2-input gate produces the output shown. (X represents the output.) This is a(n)

- a. OR gate
- b. AND gate
- c. NOR gate
- d. NAND gate



Quiz

9. A 2-input gate produces a HIGH output only when the inputs agree. This type of gate is a(n)

- a. OR gate
- b. AND gate
- c. NOR gate
- d. XNOR gate

Quiz.

10. The required logic for a PLD can be specified in an Hardware Description Language by

- a. text entry
- b. schematic entry
- c. state diagrams
- d. all of the above

Quiz

1. The associative law for addition is normally written as

a. $A + B = B + A$

b. $(A + B) + C = A + (B + C)$

c. $AB = BA$

d. $A + AB = A$

Quiz

2. The Boolean equation $AB + AC = A(B + C)$ illustrates
- a. the distribution law
 - b. the commutative law
 - c. the associative law
 - d. DeMorgan's theorem

Quiz

3. The Boolean expression $A \cdot 1$ is equal to

- a. A
- b. B
- c. 0
- d. 1

Quiz

4. The Boolean expression $A + 1$ is equal to

- a. A
- b. B
- c. 0
- d. 1

Quiz

5. The Boolean equation $AB + AC = A(B + C)$ illustrates
- a. the distribution law
 - b. the commutative law
 - c. the associative law
 - d. DeMorgan's theorem

Quiz

6. A Boolean expression that is in standard SOP form is
- a. the minimum logic expression
 - b. contains only one product term
 - c. has every variable in the domain in every term
 - d. none of the above

Quiz

7. Adjacent cells on a Karnaugh map differ from each other by
- a. one variable
 - b. two variables
 - c. three variables
 - d. answer depends on the size of the map

Quiz

8. The minimum expression that can be read from the Karnaugh map shown is

a. $X = A$

b. $X = \bar{A}$

c. $X = B$

d. $X = \bar{B}$

	$\bar{C}$	C
$\bar{A}\bar{B}$		
$\bar{A}B$		
AB	1	1
$A\bar{B}$	1	1

Quiz

9. The minimum expression that can be read from the Karnaugh map shown is

a. $X = A$

b. $X = \bar{A}$

c. $X = B$

d. $X = \bar{B}$

	$\bar{C}$	C
$\bar{A}\bar{B}$	1	1
$\bar{A}B$		
AB		
$A\bar{B}$	1	1

Quiz

10. In VHDL code, the two main parts are called the

- a. I/O and the module
- b. entity and the architecture
- c. port and the module
- d. port and the architecture

Quiz

1. Assume an AOI expression is $\overline{AB + CD}$. The equivalent POS expression is

a. $(A + B)(C + D)$

b. $(\overline{A} + \overline{B})(\overline{C} + \overline{D})$

c. $\overline{(A + B)}\overline{(C + D)}$

d. none of the above

Quiz.

2. The truth table shown is for

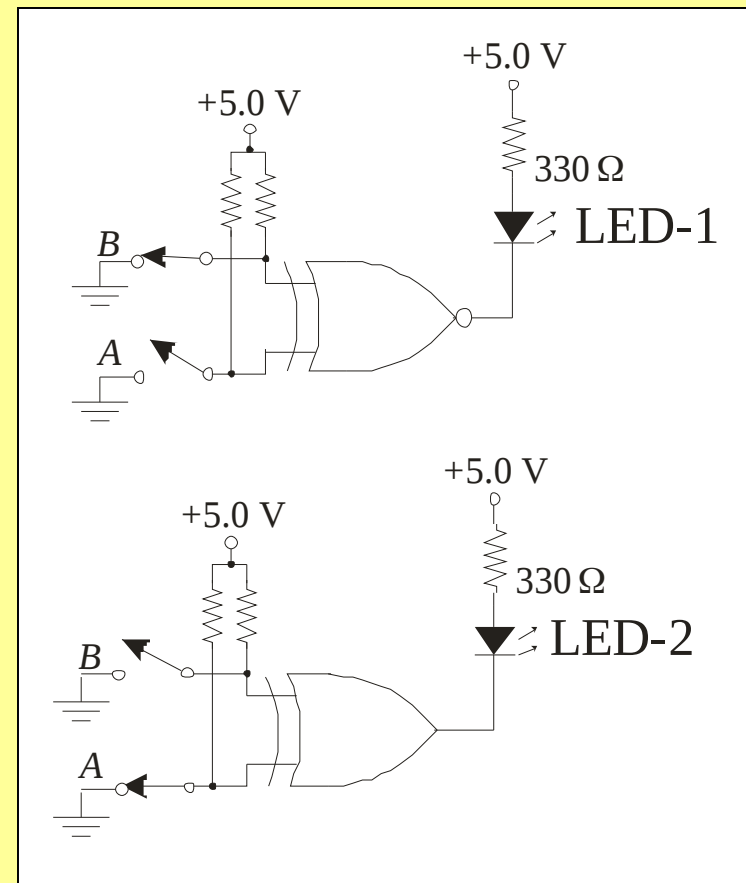
- a. a NAND gate
- b. a NOR gate
- c. an exclusive-OR gate
- d. an exclusive-NOR gate

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

Quiz

3. An LED that should be ON is

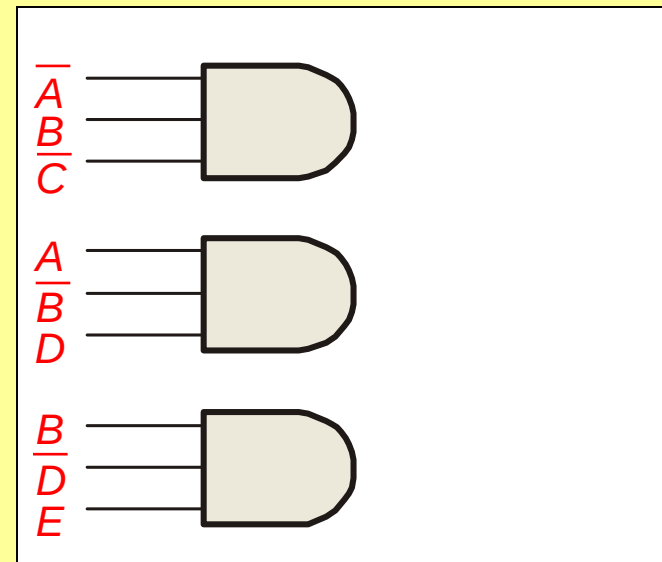
- a. LED-1
- b. LED-2
- c. neither
- d. both



Quiz

4. To implement the SOP expression $X = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}D + B\bar{D}E$, the type of gate that is needed is a

- a. 3-input AND gate
- b. 3-input NAND gate
- c. 3-input OR gate
- d. 3-input NOR gate



Quiz

5. Reading the Karnaugh map, the logic expression is

a. $\overline{A}\overline{C} + \overline{A}B$

b. $\overline{A}B + A\overline{C}$

c. $A\overline{B} + B\overline{C}$

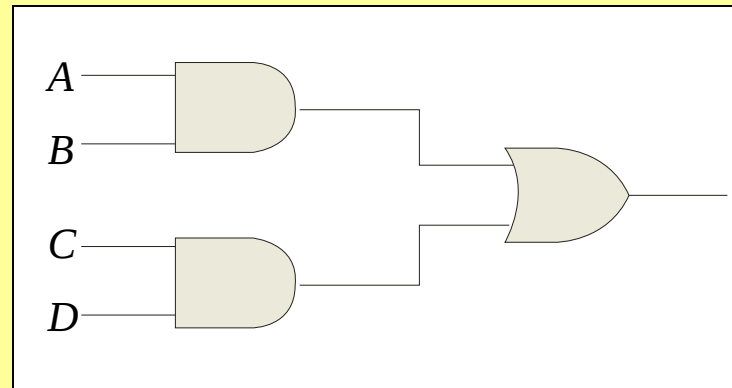
d. $\overline{A}B + \overline{A}\overline{C}$

	$\overline{C}$	C
$\overline{A}\overline{B}$	1	
$\overline{A}B$	1	1
AB		
$A\overline{B}$		

Quiz

6. The circuit shown will have identical logic out if all gates are changed to

- a. AND gates
- b. OR gates
- c. NAND gates
- d. NOR gates



Quiz.

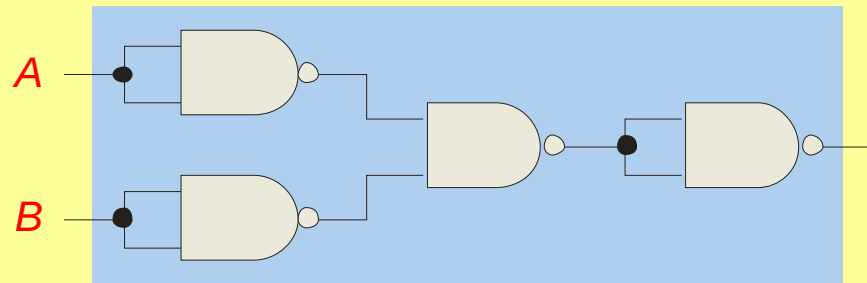
7. The two types of gates which are called *universal gates* are

- a. AND/OR
- b. NAND/NOR
- c. AND/NAND
- d. OR/NOR

Quiz

8. The circuit shown is equivalent to an

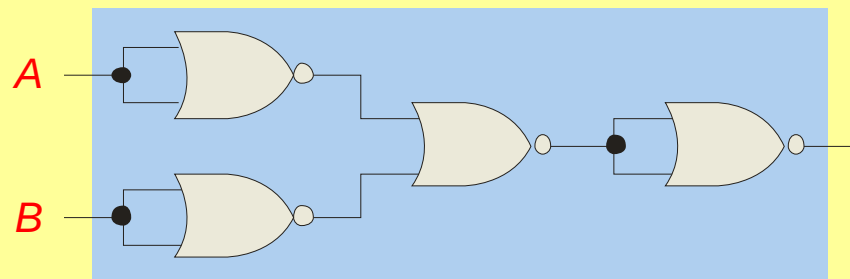
- a. AND gate
- b. XOR gate
- c. OR gate
- d. none of the above



Quiz

9. The circuit shown is equivalent to

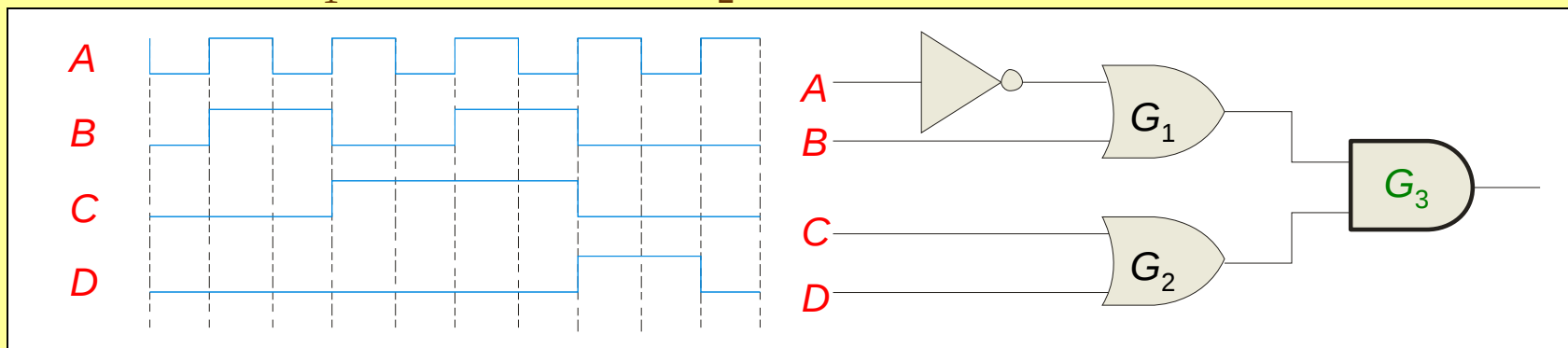
- a. an AND gate
- b. an XOR gate
- c. an OR gate
- d. none of the above



Quiz

10. During the first *three* intervals for the pulsed circuit shown, the output of

- a. G_1 is LOW and G_2 is LOW
- b. G_1 is LOW and G_2 is HIGH
- c. G_1 is HIGH and G_2 is LOW
- d. G_1 is HIGH and G_2 is HIGH



Quiz

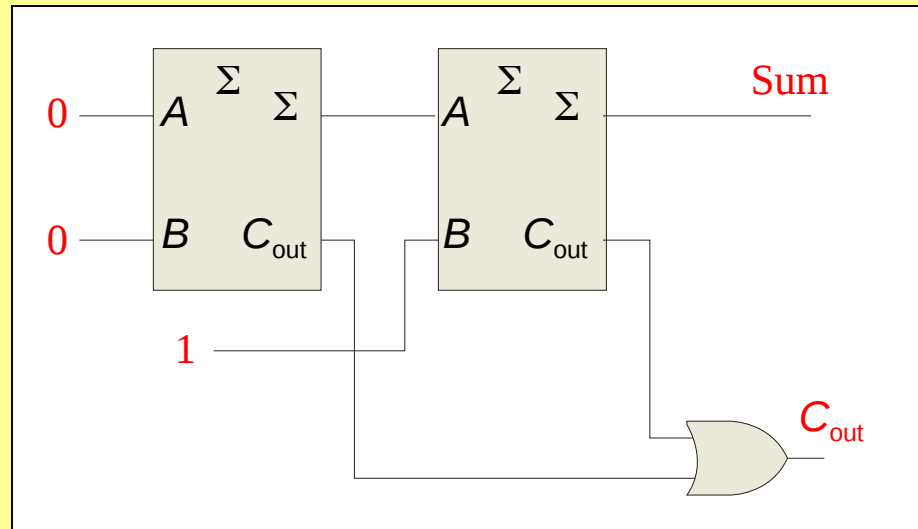
1. For the full-adder shown, assume the input bits are as shown with $A = 0$, $B = 0$, $C_{in} = 1$. The **Sum** and C_{out} will be

a. $\text{Sum} = 0$ $C_{out} = 0$

b. $\text{Sum} = 0$ $C_{out} = 1$

c. $\text{Sum} = 1$ $C_{out} = 0$

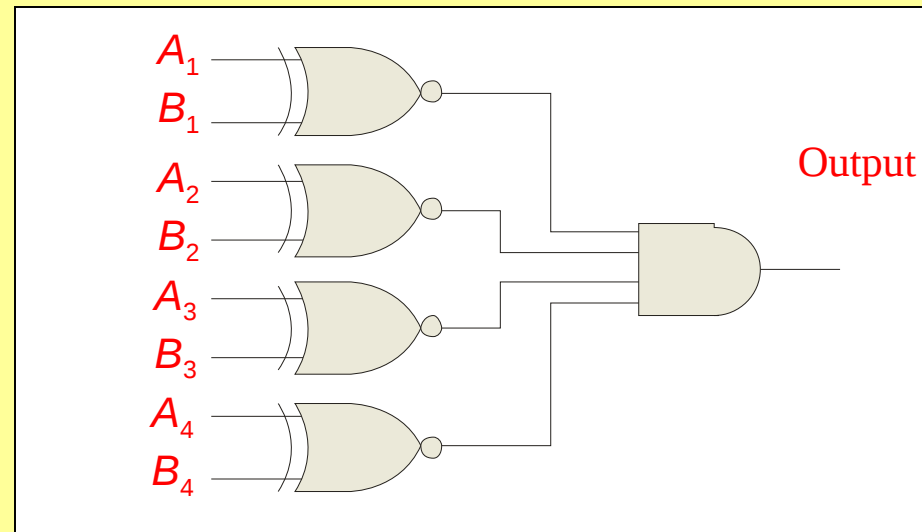
d. $\text{Sum} = 1$ $C_{out} = 1$



Quiz

2. The output will be LOW if

- a. $A < B$
- b. $A > B$
- c. both a and b are correct
- d. $A = B$



Quiz

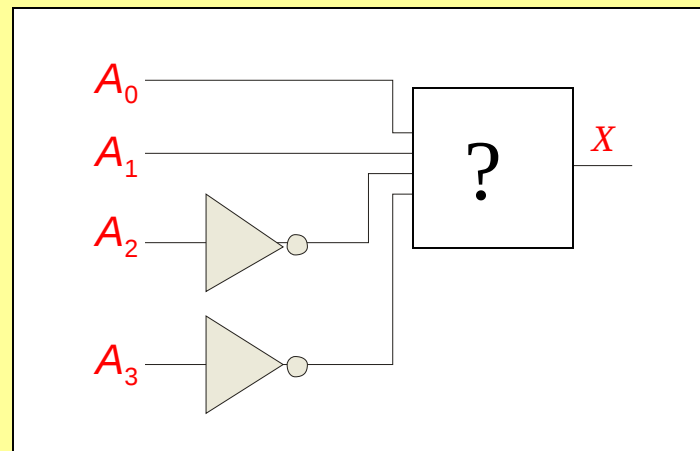
3. If you expand two 4-bit comparators to accept two 8-bit numbers, the output of the least significant comparator is

- a. equal to the final output
- b. connected to the cascading inputs of the most significant comparator
- c. connected to the output of the most significant comparator
- d. not used

Quiz

4. Assume you want to decode the binary number 0011 with an active-LOW decoder. The missing gate should be

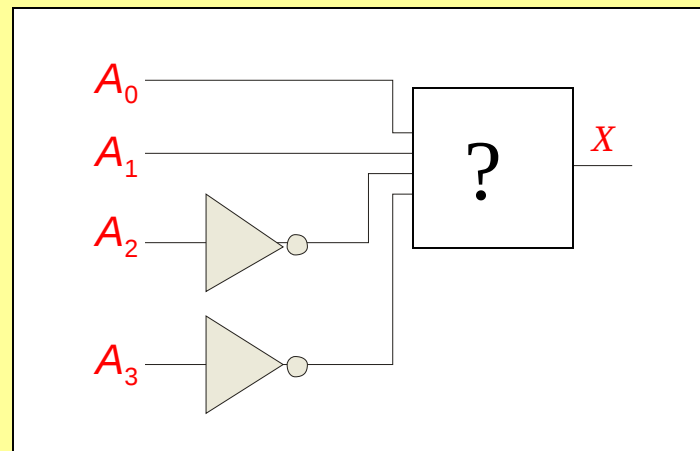
- a. an AND gate
- b. an OR gate
- c. a NAND gate
- d. a NOR gate



Quiz

5. Assume you want to decode the binary number 0011 with an active-HIGH decoder. The missing gate should be

- a. an AND gate
- b. an OR gate
- c. a NAND gate
- d. a NOR gate



Quiz

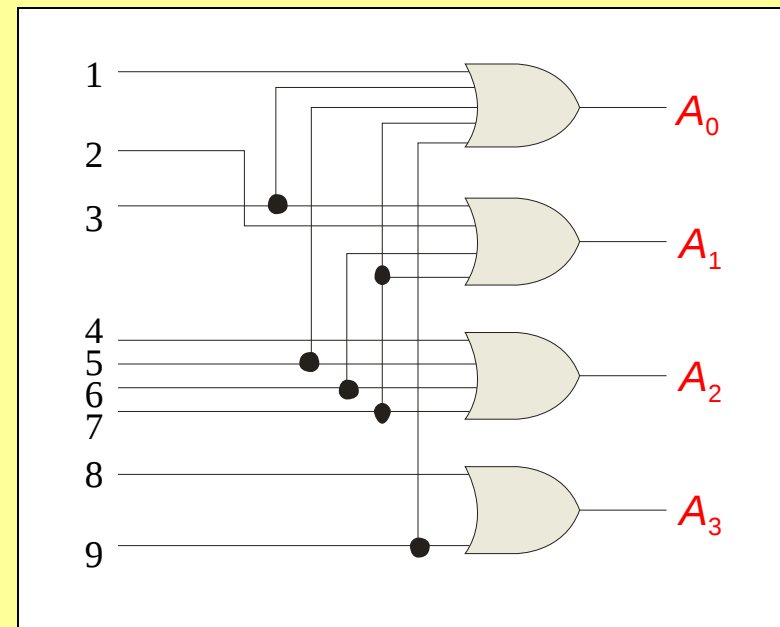
6. The 74138 is a 3-to-8 decoder. Together, two of these ICs can be used to form one 4-to-16 decoder. To do this, connect

- a. one decoder to the LSBs of the input; the other decoder to the MSBs of the input
- b. all chip select lines to ground
- c. all chip select lines to their active levels
- d. one chip select line on each decoder to the input MSB

Quiz

7. The decimal-to-binary encoder shown does not have a zero input. This is because

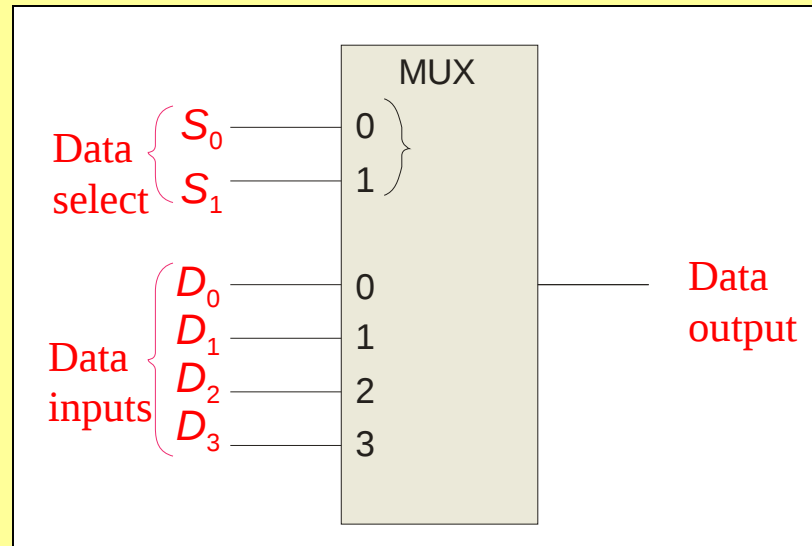
- a. when zero is the input, all lines should be LOW
- b. zero is not important
- c. zero will produce illegal logic levels
- d. another encoder is used for zero



Quiz

8. If the data select lines of the MUX are $S_1S_0 = 11$, the output will be

- a. LOW
- b. HIGH
- c. equal to D_0
- d. equal to D_3



Quiz

9. The 74138 decoder can also be used as
- a. an encoder
 - b. a DEMUX
 - c. a MUX
 - d. none of the above

Quiz

10. The 74LS280 can generate even or odd parity. It can also be used as

- a. an adder
- b. a parity tester
- c. a MUX
- d. an encoder

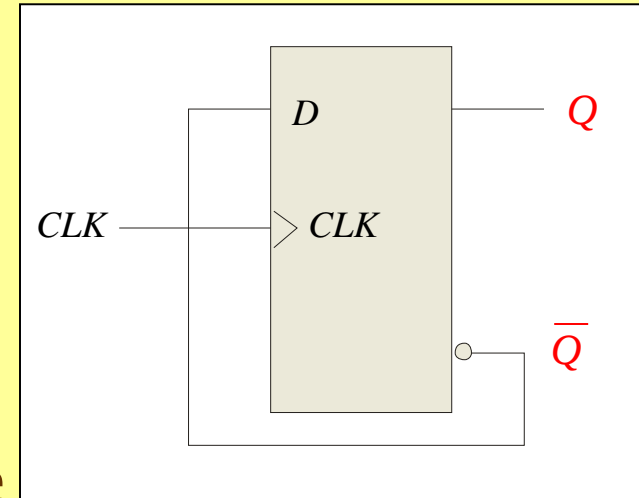
Quiz.

1. The output of a D latch will not change if
 - a. the output is LOW
 - b. Enable is not active
 - c. D is LOW
 - d. all of the above

Quiz

2. The D flip-flop shown will

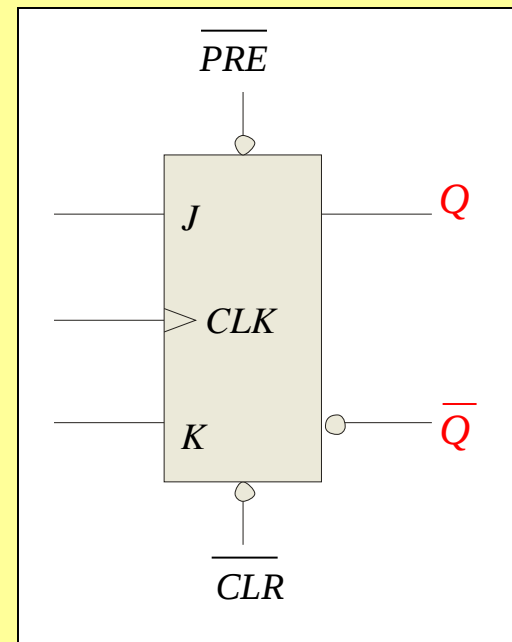
- a. set on the next clock pulse
- b. reset on the next clock pulse
- c. latch on the next clock pulse
- d. toggle on the next clock pulse



Quiz

3. For the J-K flip-flop shown, the number of inputs that are asynchronous is

- a. 1
- b. 2
- c. 3
- d. 4



Quiz

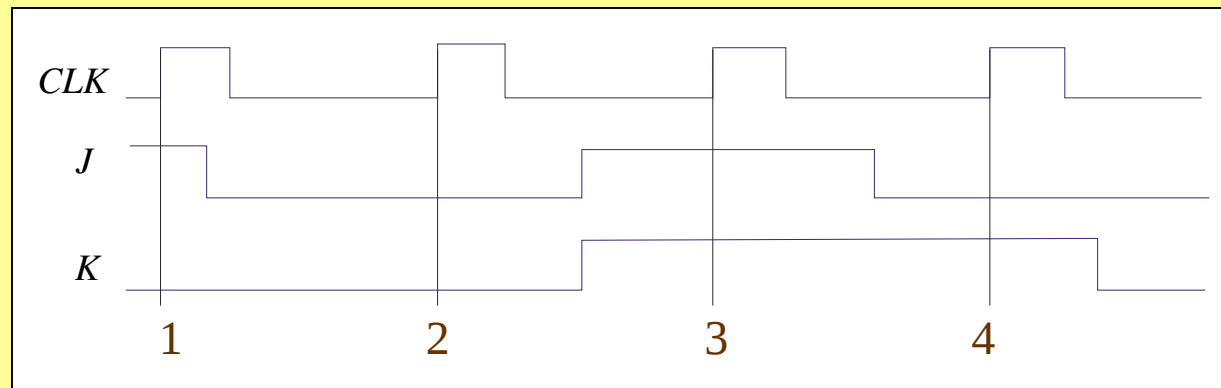
4. Assume the output is initially HIGH on a leading edge triggered J-K flip flop. For the inputs shown, the output will go from HIGH to LOW on which clock pulse?

a. 1

b. 2

c. 3

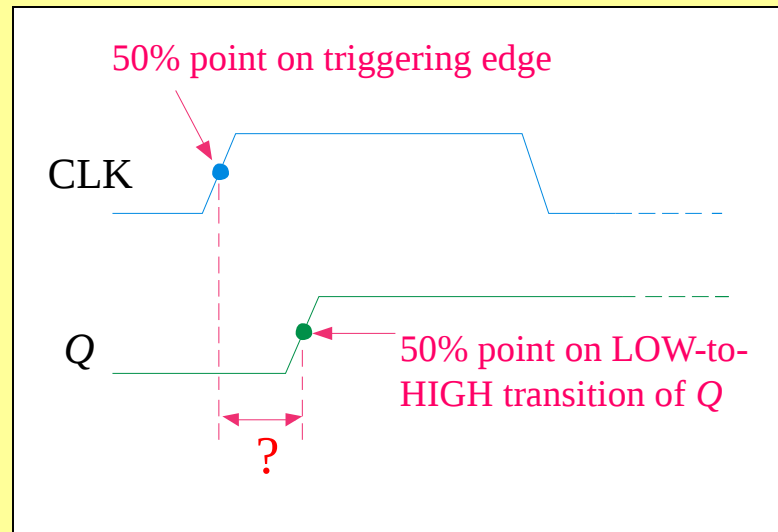
d. 4



Quiz

5. The time interval illustrated is called

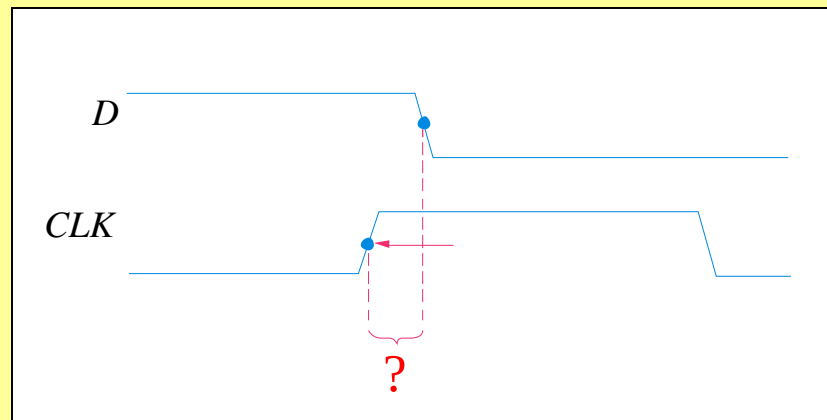
- a. t_{PHL}
- b. t_{PLH}
- c. set-up time
- d. hold time



Quiz.

6. The time interval illustrated is called

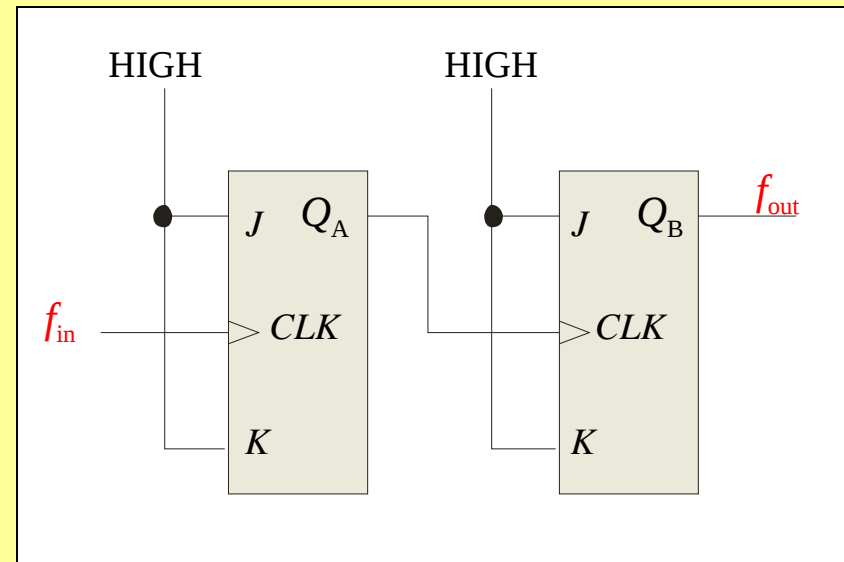
- a. t_{PHL}
- b. t_{PLH}
- c. set-up time
- d. hold time



Quiz

7. The application illustrated is a

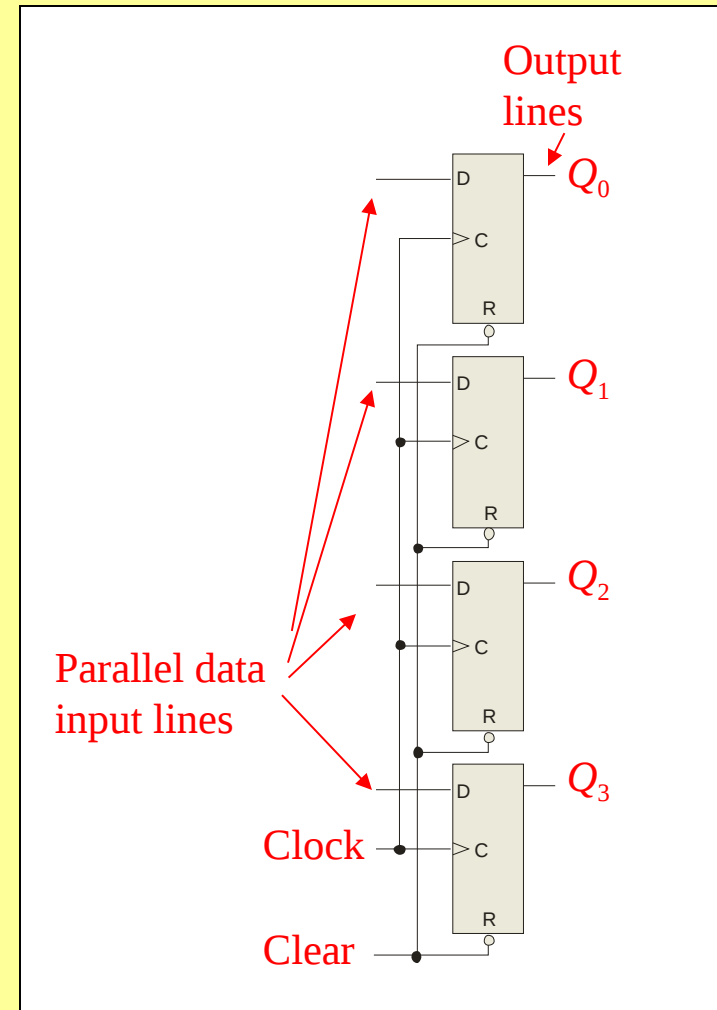
- a. astable multivibrator
- b. data storage device
- c. frequency multiplier
- d. frequency divider



Quiz

8. The application illustrated is a

- a. astable multivibrator
- b. data storage device
- c. frequency multiplier
- d. frequency divider



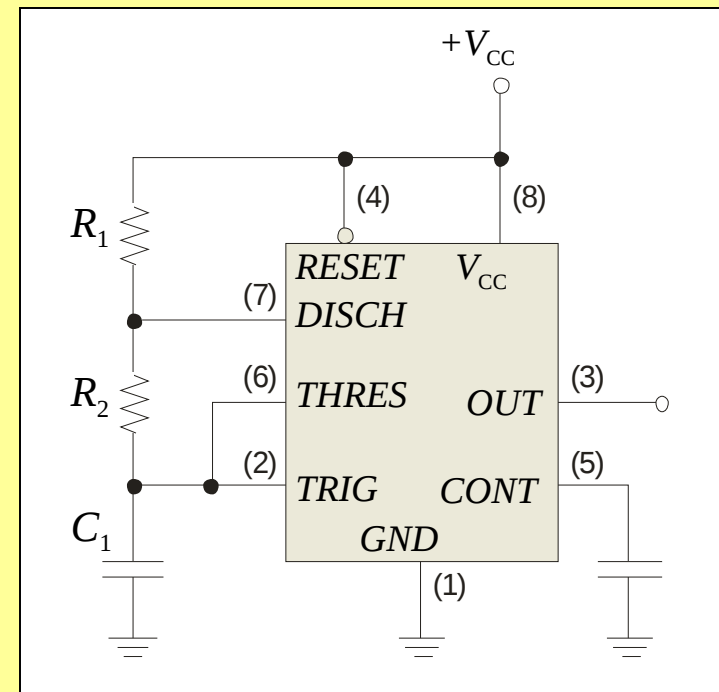
Quiz

9. A retriggerable one-shot with an active HIGH output has a pulse width of 20 ms and is triggered from a 60 Hz line. The output will be a

- a. series of 16.7 ms pulses
- b. series of 20 ms pulses
- c. constant LOW
- d. constant HIGH

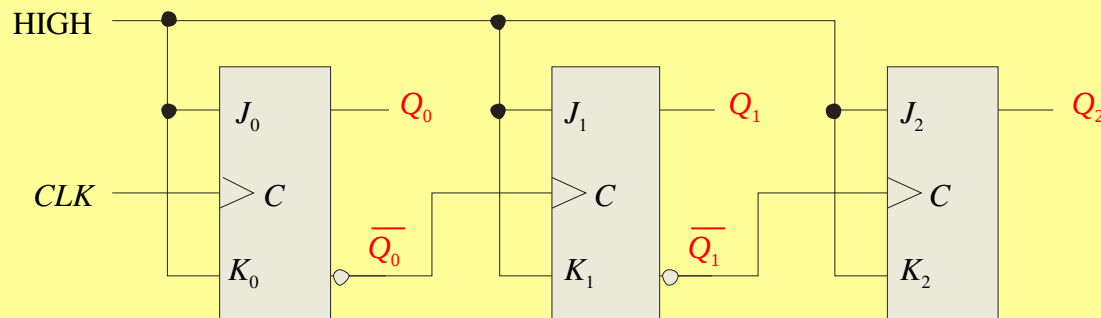
Quiz

10. The circuit illustrated is a
- a. astable multivibrator
 - b. monostable multivibrator
 - c. frequency multiplier
 - d. frequency divider



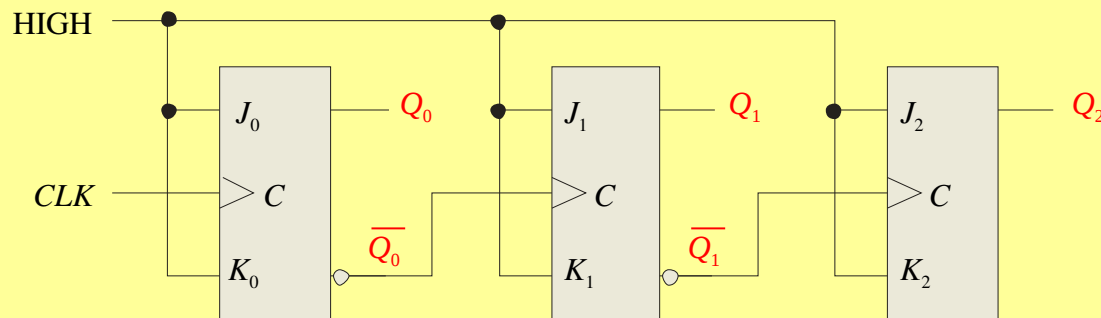
Quiz

1. The counter shown below is an example of
- a. an asynchronous counter
 - b. a BCD counter
 - c. a synchronous counter
 - d. none of the above



Quiz

2. The Q_0 output of the counter shown
- a. is present before Q_1 or Q_2
 - b. changes on every clock pulse
 - c. has a higher frequency than Q_1 or Q_2
 - d. all of the above



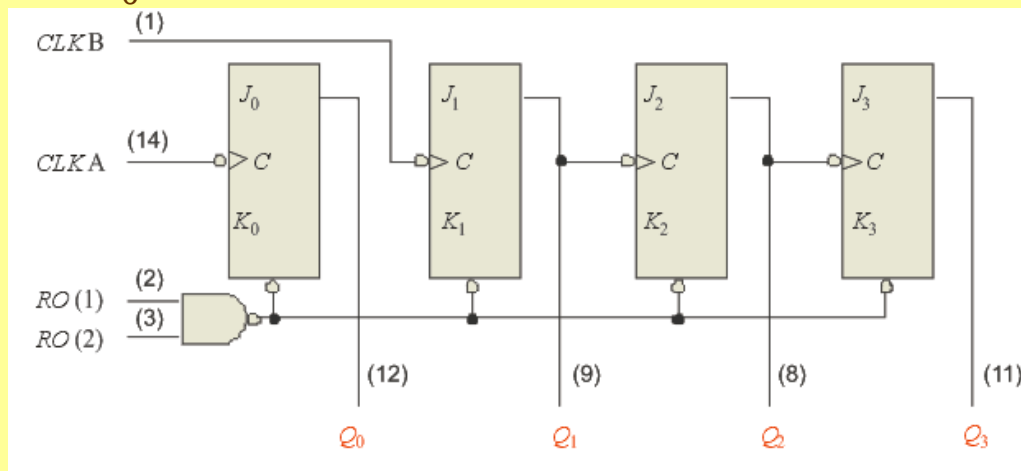
Quiz

3. To cause a D flip-flop to toggle, connect the
- a. clock to the D input
 - b. Q output to the D input
 - c. $\overline{Q}$ output to the D input
 - d. clock to the preset input

Quiz

4. The 7493A asynchronous counter diagram is shown (J 's and K 's are HIGH.) To make the count have a modulus of 16, connect

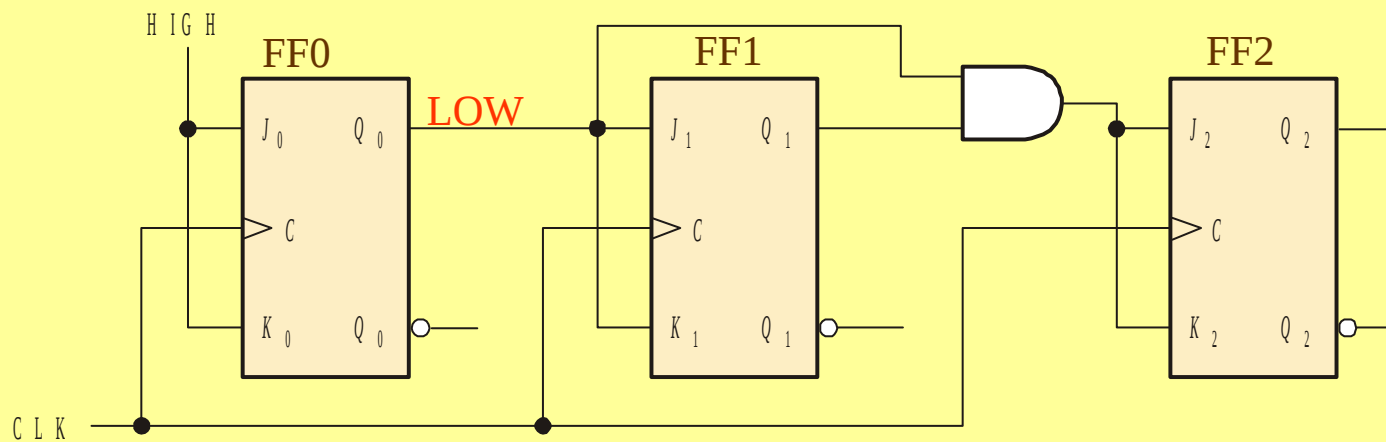
- a. Q_0 to RO(1) and RO(2) to
- b. Q_3 to RO(1) and RO(2)
- c. $CLK A$ and $CLK B$ together
- d. Q_0 to $CLK B$



Quiz.

5. Assume Q_0 is LOW. The next clock pulse will cause

- a. FF1 and FF2 to both toggle
- b. FF1 and FF2 to both latch
- c. FF1 to latch; FF2 to toggle
- d. FF1 to toggle; FF2 to latch



Quiz

6. A 4-bit binary counter has a terminal count of
- a. 4
 - b. 10
 - c. 15
 - d. 16

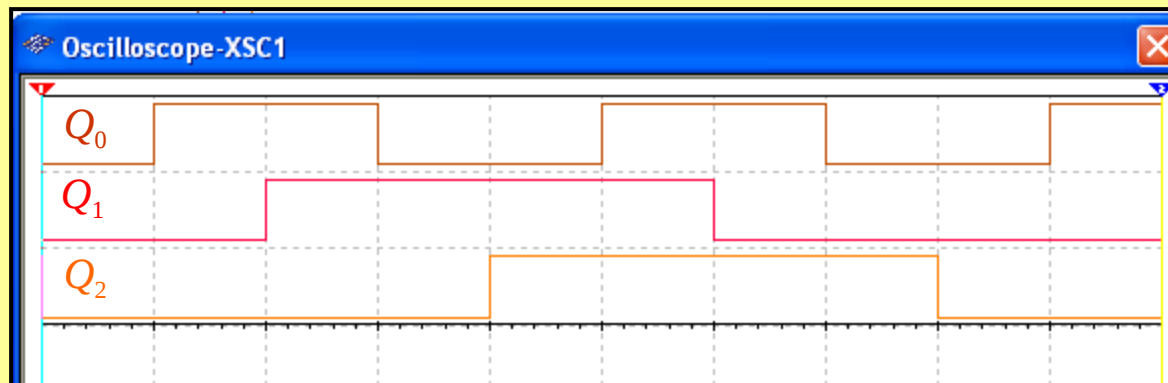
Quiz

7. Assume the clock for a 4-bit binary counter is 80 kHz. The output frequency of the fourth stage (Q_3) is
- a. 5 kHz
 - b. 10 kHz
 - c. 20 kHz
 - d. 320 kHz

Quiz

8. A 3-bit count sequence is shown for a counter (Q_2 is the MSB). The sequence is

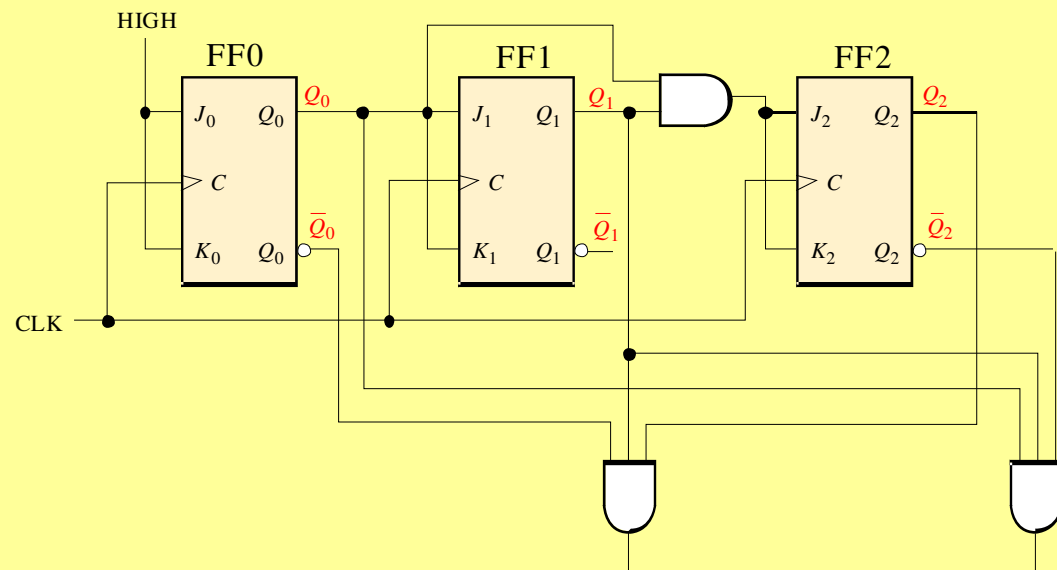
- a. 0-1-2-3-4-5-6-7-0 (repeat)
- b. 0-1-3-2-6-7-5-4-0 (repeat)
- c. 0-2-4-6-1-3-5-7-0 (repeat)
- d. 0-4-6-2-3-7-5-1-0 (repeat)



Quiz.

9. FF2 represents the MSB. The counts that are being decoded by the 3-input AND gates are

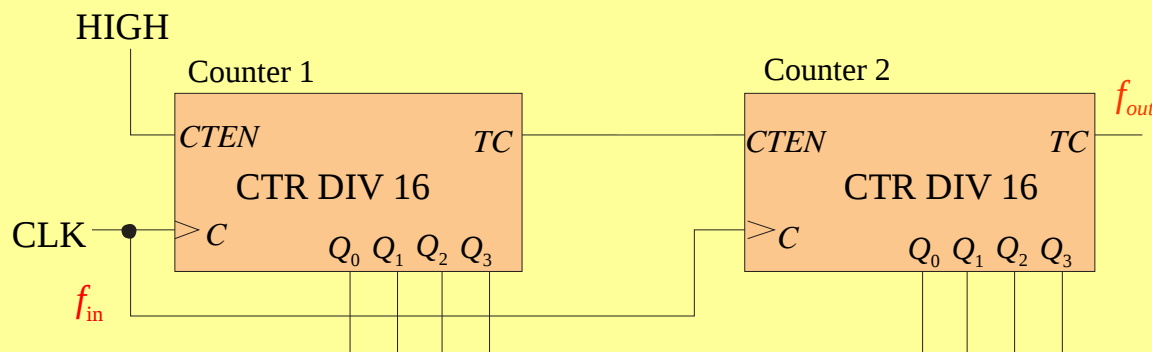
- a. 2 and 3
- b. 3 and 6
- c. 2 and 5
- d. 5 and 6



Quiz

10. Assume the input frequency (f_{in}) is 256 Hz. The output frequency (f_{out}) will be

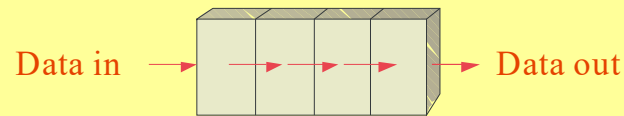
- a. 16 Hz
- b. 1 kHz
- c. 65 kHz
- d. none of the above



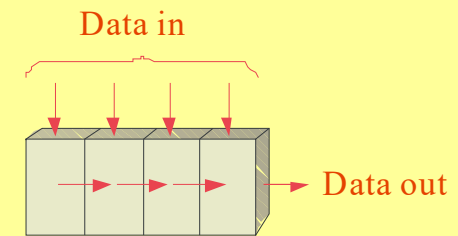
Quiz

1. The shift register that would be used to delay serial data by 4 clock periods is

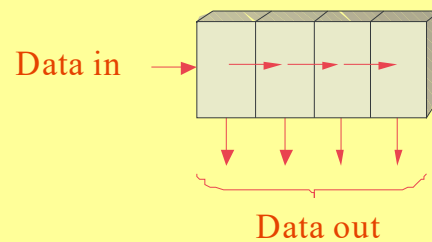
a.



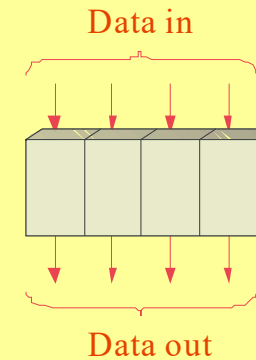
c.



b.

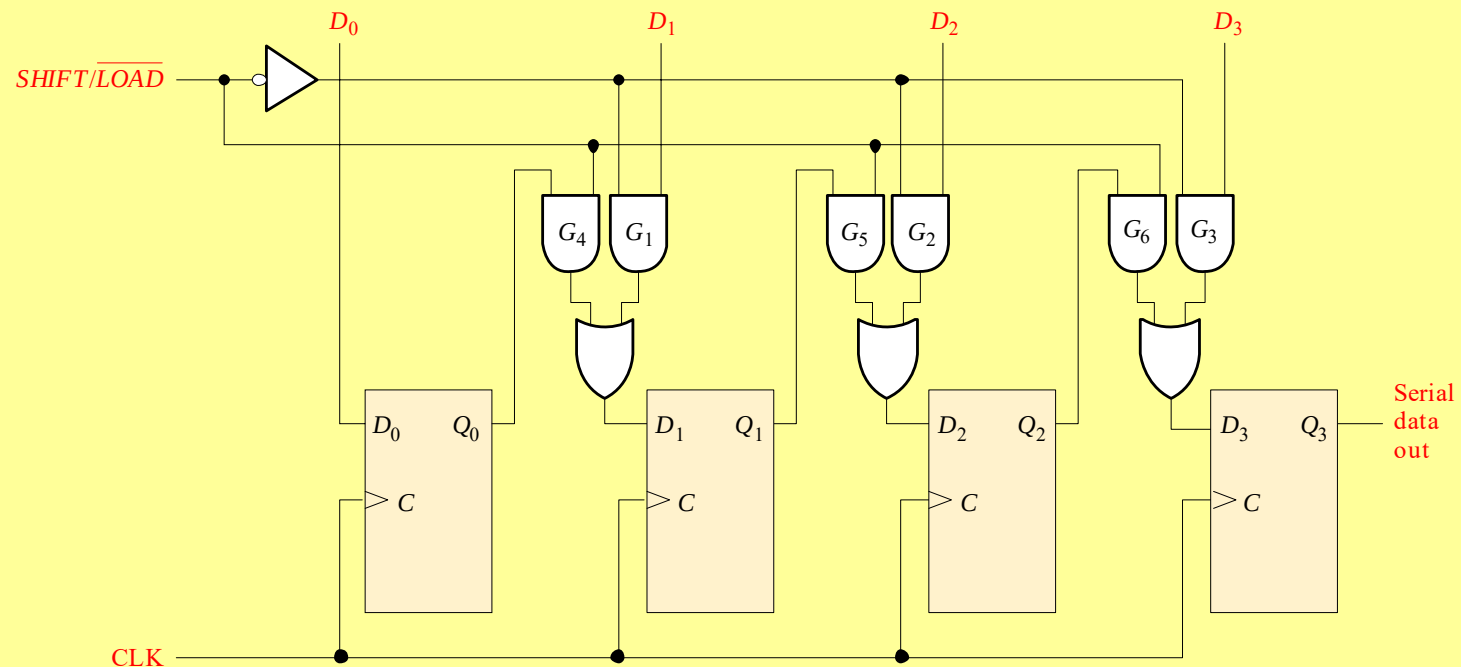


d.



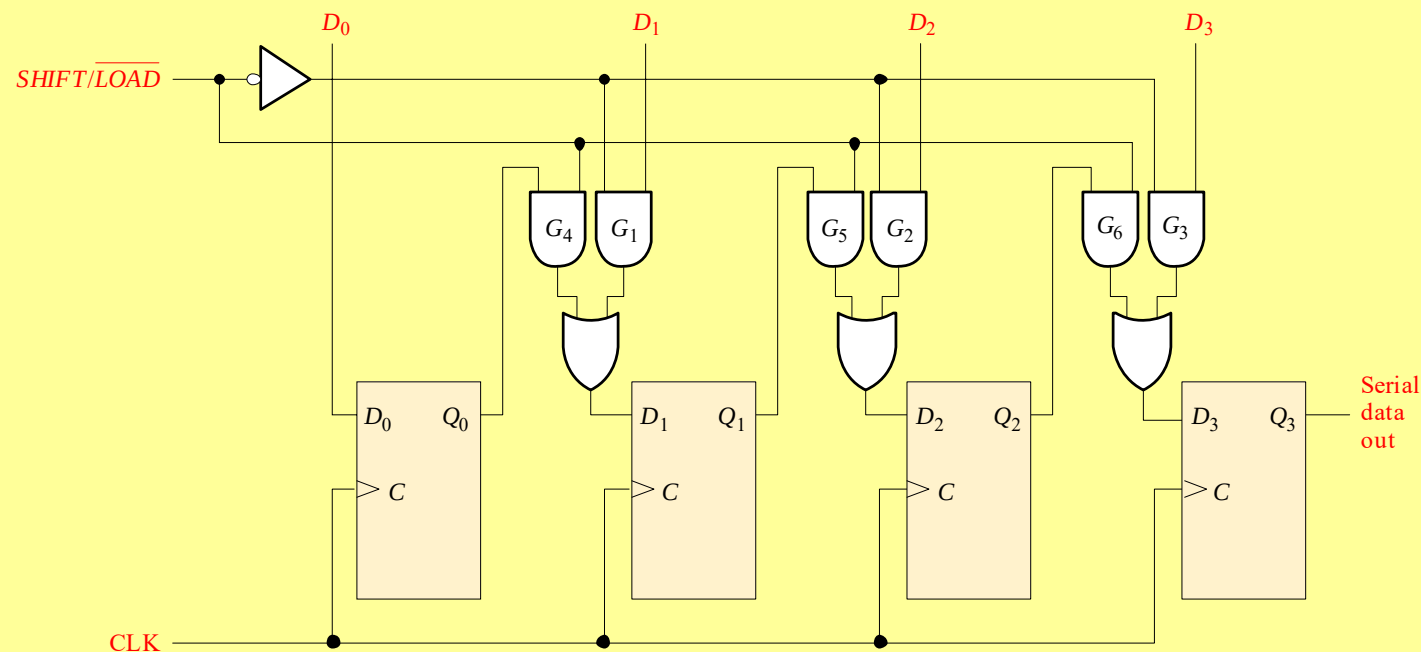
Quiz

2. The circuit shown is a
- a. serial-in/serial-out shift register
 - b. serial-in/parallel-out shift register
 - c. parallel-in/serial-out shift register
 - d. parallel-in/parallel-out shift register



Quiz

3. If the $\overline{SHIFT/LOAD}$ line is HIGH, data
- a. is loaded from D_0 , D_1 , D_2 and D_3 immediately
 - b. is loaded from D_0 , D_1 , D_2 and D_3 on the next CLK
 - c. shifted from left to right on the next CLK
 - d. shifted from right to left on the next CLK



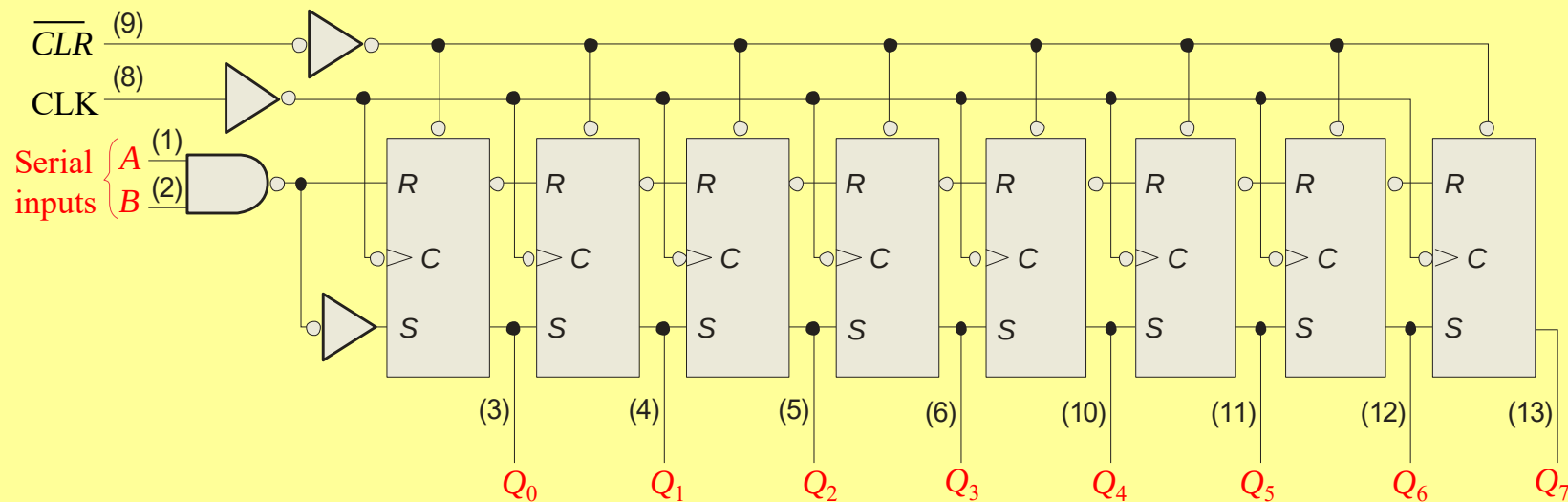
Quiz

4. A 4-bit parallel-in/parallel-out shift register will store data for
- a. 1 clock period
 - b. 2 clock periods
 - c. 3 clock periods
 - d. 4 clock periods

Quiz

5. The 74HC164 (shown) has two serial inputs. If data is placed on the *A* input, the *B* input

- a. could serve as an active LOW enable
- b. could serve as an active HIGH enable
- c. should be connected to ground
- d. should be left open



Quiz

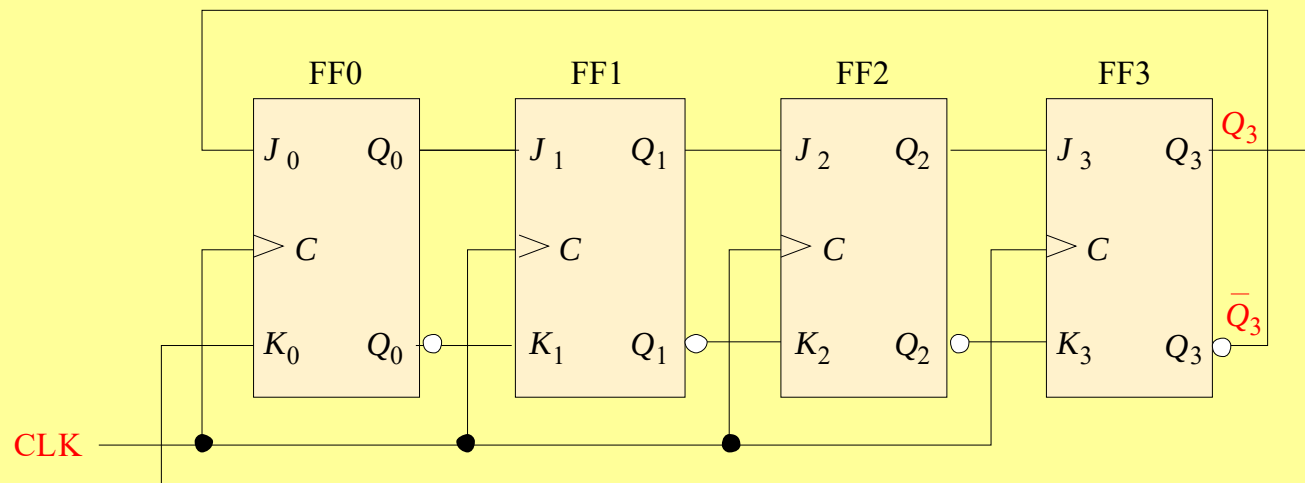
6. An advantage of a ring counter over a Johnson counter is that the ring counter
- a. has more possible states for a given number of flip-flops
 - b. is cleared after each cycle
 - c. allows only one bit to change at a time
 - d. is self-decoding

Quiz

7. A possible sequence for a 4-bit ring counter is
- a. ... 1111, 1110, 1101 ...
 - b. ... 0000, 0001, 0010 ...
 - c. ... 0001, 0011, 0111 ...
 - d. ... 1000, 0100, 0010 ...

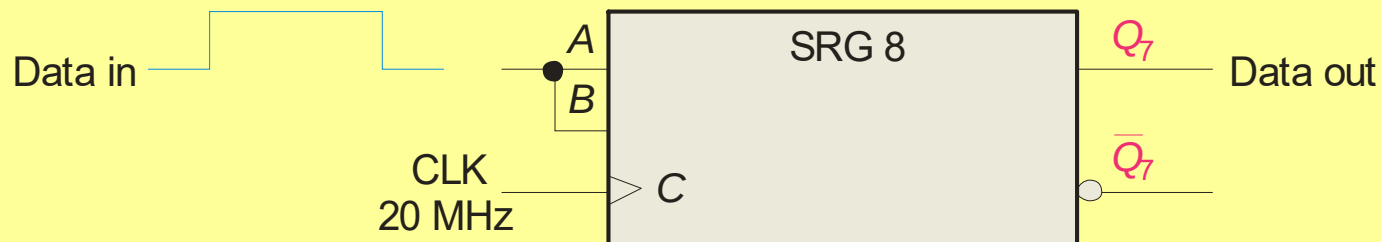
Quiz

8. The circuit shown is a
- a. serial-in/parallel-out shift register
 - b. serial-in/serial-out shift register
 - c. ring counter
 - d. Johnson counter



Quiz

9. Assume serial data is applied to the 8-bit shift register shown. The clock frequency is 20 MHz. The first data bit will show up at the output in
- a. 50 ns
 - b. 200 ns
 - c. 400 ns
 - d. 800 ns



Quiz

10. For transmission, data from a UART is sent in

- a. asynchronous serial form
- b. synchronous parallel form
- c. can be either of the above
- d. none of the above

Quiz

1. Static RAM is

- a. nonvolatile read only memory
- b. nonvolatile read/write memory
- c. volatile read only memory
- d. volatile read/write memory

Quiz

2. A nonvolatile memory is one that
- a. requires a clock
 - b. must be refreshed regularly
 - c. retains data without power applied
 - d. all of the above

Quiz

3. The advantage of dynamic RAM over static RAM is that
- a. it is much faster
 - b. it does not require refreshing
 - c. it is simpler and cheaper
 - d. all of the above

Quiz

4. The first step in a read or write operation for a random access memory is to

- a. place a valid address on the address bus
- b. enable the memory
- c. send or obtain the data
- d. start a refresh cycle

Quiz

5. The output enable signal ($\overline{OE}$) on a RAM is active
- a. only during a write operation
 - b. only during a read operation
 - c. both of the above
 - d. none of the above

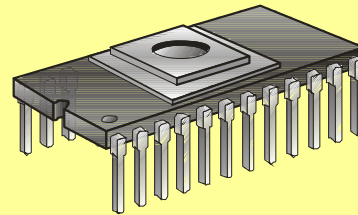
Quiz

6. When data is read from RAM, the memory location is
- a. cleared after the read operation
 - b. set to all 1's after the read operation
 - c. unchanged
 - d. destroyed

Quiz

7. An EPROM has a window to allow UV light to enter under certain conditions. The purpose of this is to

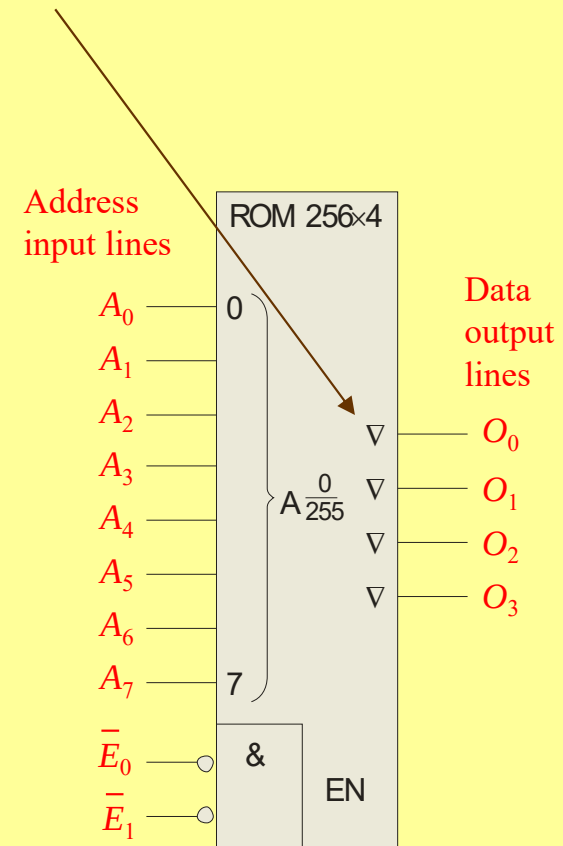
- a. refresh the data
- b. read the data
- c. program the IC
- d. erase the data



Quiz

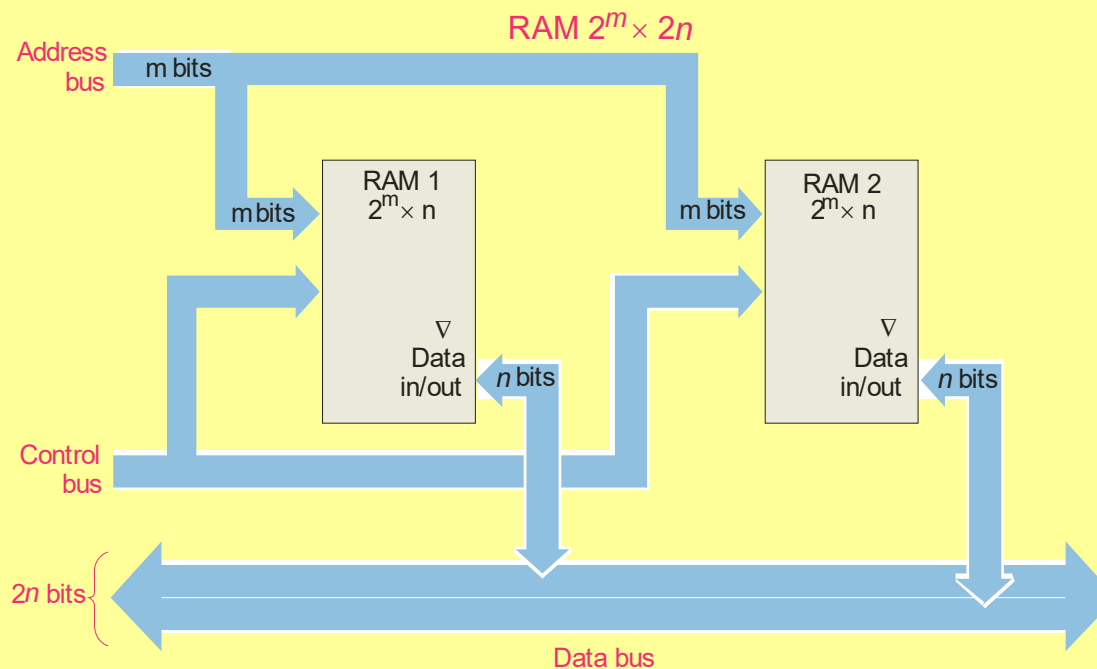
8. The small triangles on the logic diagram indicate that these outputs are

- a. not used
- b. tri-stated
- c. inverted
- d. grounded



Quiz

9. Using two ICs as shown will expand
- a. the word size
 - b. the number of words available
 - c. both of the above
 - d. none of the above



Quiz

10. On a hard drive, information about file names, locations, and file size are kept in a special location called the

- a. file location list
- b. file allocation table
- c. disk directory
- d. stack

Quiz

1. An advantage of PLDs over discrete circuits is
 - a. lower power and space requirements
 - b. higher reliability
 - c. design flexibility
 - d. all of the above

Quiz

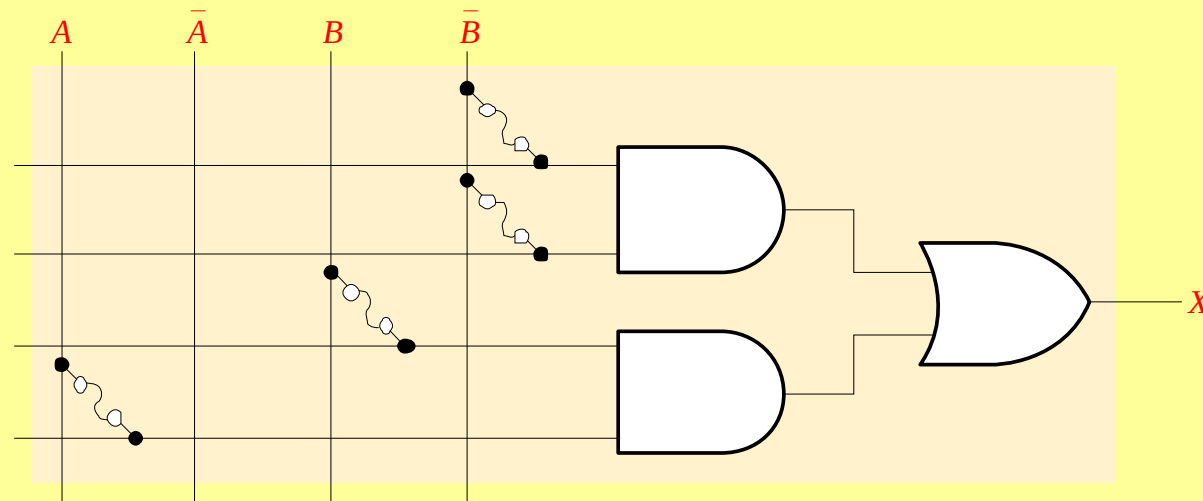
2. The logic expression for X is

a. $X = \bar{B}(A + B)$

b. $X = \bar{B} + A\bar{B}$

c. $X = \bar{B} + AB$

d. $X = B(A + \bar{B})$



Quiz

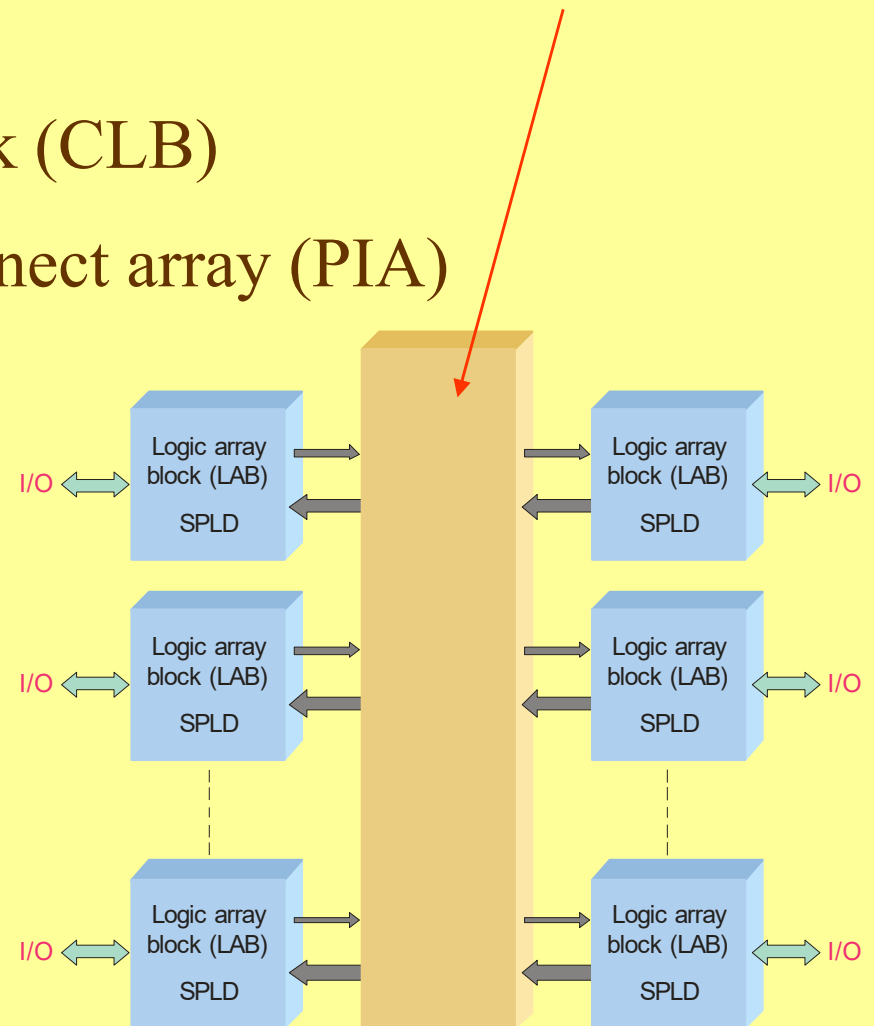
3. Generic Array Logic (GAL)

- a. is reprogrammable
- b. uses look-up tables for combinational logic
- c. uses SRAM technology
- d. all of the above

Quiz

4. A general block of a CPLD is shown. The center (unmarked) block represents a

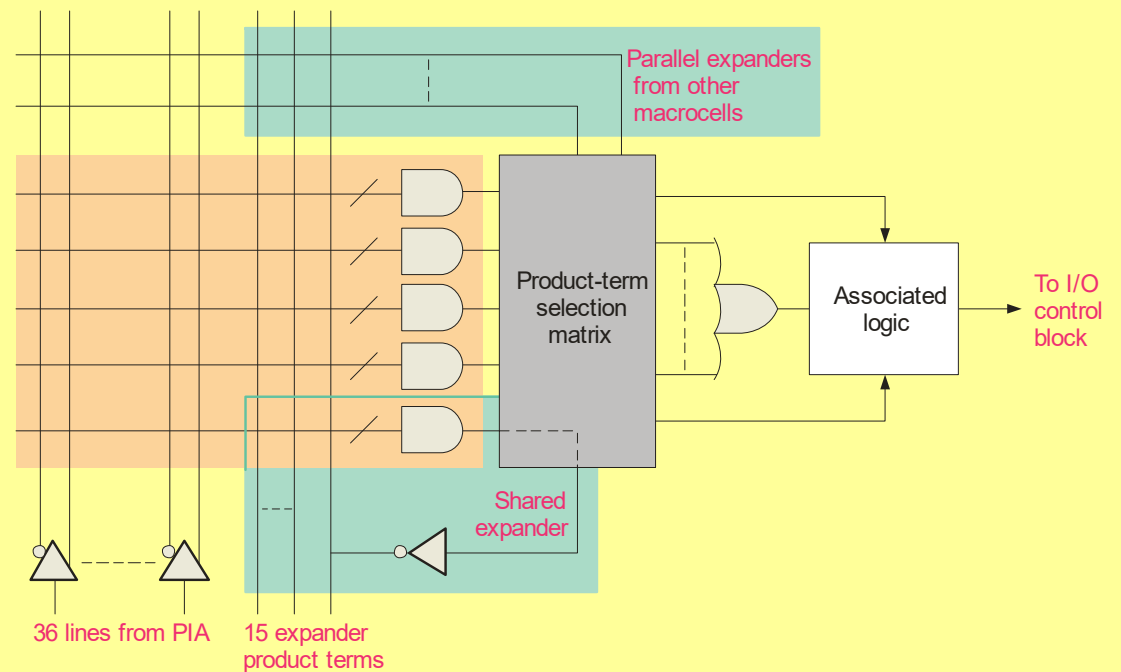
- a. configurable logic block (CLB)
- b. programmable interconnect array (PIA)
- c. comparator
- d. look-up table (LUT)



Quiz

5. The diagram represents

- a. a PIA
- b. an FPGA
- c. a logic module
- d. a macrocell



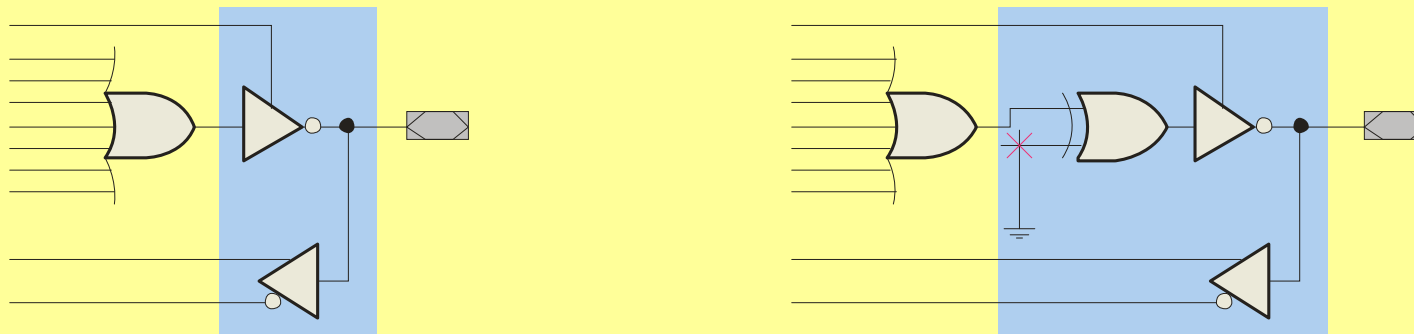
Quiz

6. A programmable device that uses a LUT to generate logic is

- a. a PAL
- b. a GAL
- c. an FPGA
- d. a CPLD

Quiz

7. The drawings represent two types of
- a. expanders
 - b. macrocells
 - c. logic array blocks
 - d. sequential logic blocks



Quiz

8. VHDL is a

- a. type of FPGA
- b. system programming language
- c. development software
- d. hardware description language (HDL)

Quiz

9. A written description of all of the components and connections in a circuit is called a

- a. netlist
- b. look-up table
- c. logic array list
- d. simulation table

Quiz

10. An statement in VHDL is: `QNot <= not B or not Q;`.

The `<=` characters cause

- a. variable on the left to be complemented
- b. expression on the right to be assigned to the variable on the left
- c. variable on the left to be assigned the smaller of two values
- d. constant on the left to be assigned to the expression on the right

Quiz

1. If an anti-aliasing filter is not used in digitizing a signal the recovery process

- a. is slowed
- b. may include alias signals
- c. will have less noise
- d. all of the above

Quiz

2. An anti-aliasing filter should have

- a. f_c more than 2 times the Nyquist frequency
- b. f_c equal to the Nyquist frequency
- c. f_c more than $\frac{1}{2} f_{\text{sample}}$
- d. f_c less than $\frac{1}{2} f_{\text{sample}}$

Quiz

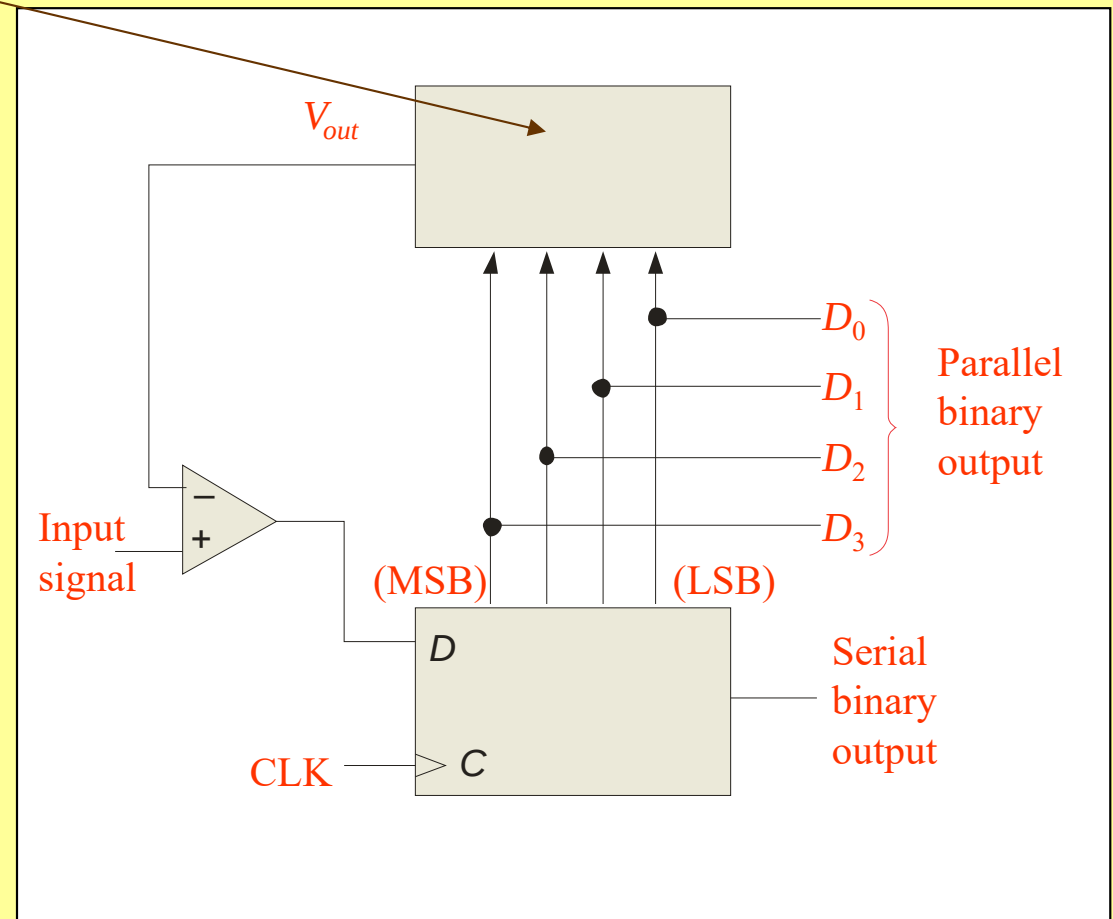
3. The number of comparators required in a 10-bit flash ADC is

- a. 255
- b. 511
- c. 1023
- d. 4095

Quiz

4. The block diagram is for a successive-approximation ADC. The top block is

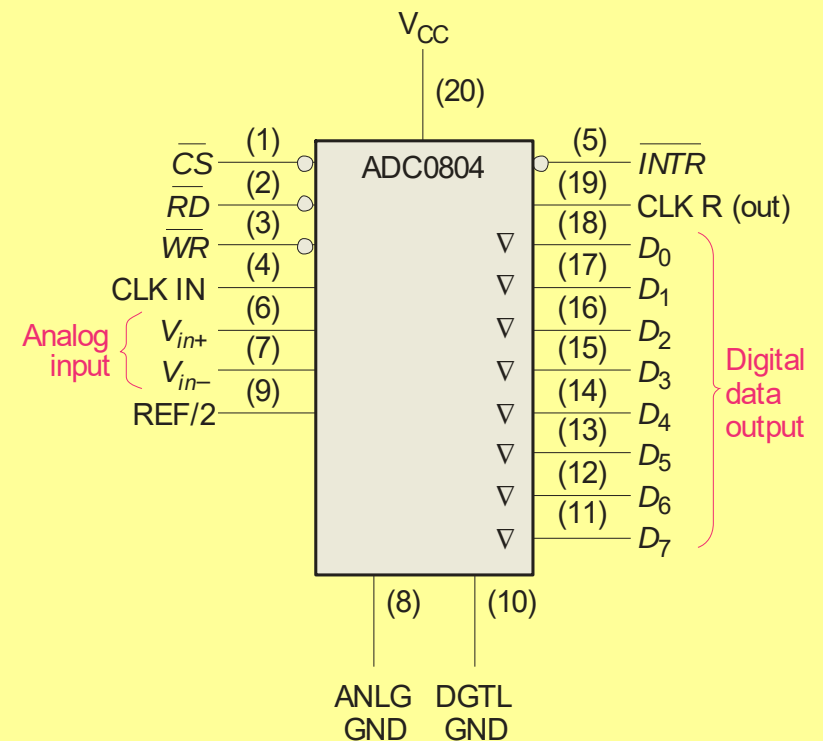
- a. an SAR
- b. a DAC
- c. an ADC
- d. a comparator



Quiz

5. The ADC0804 integrated circuit signals a completed conversion by

- a. $\overline{INTR}$ goes LOW
- b. $\overline{CS}$ goes LOW
- c. $\overline{RD}$ goes LOW
- d. CLK R goes HIGH



Quiz

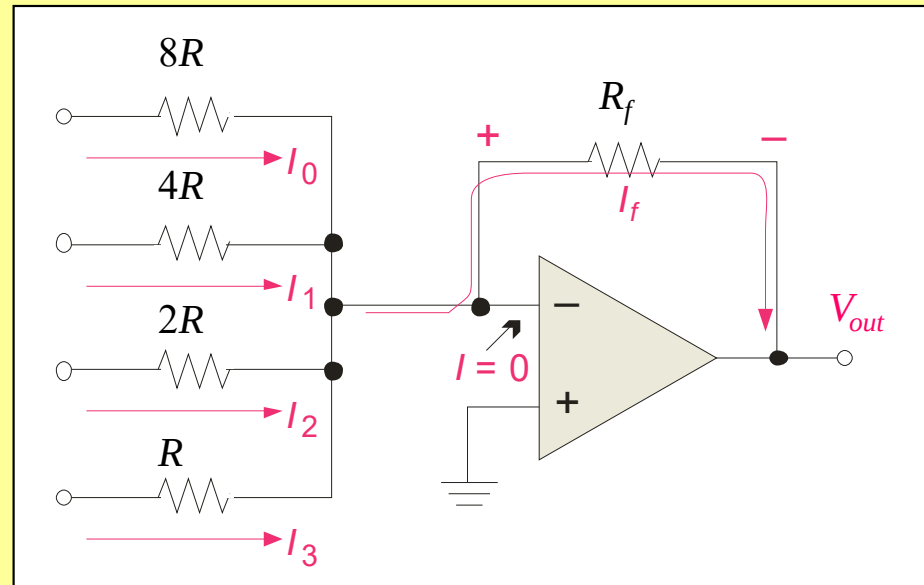
6. A sigma-delta circuit is a form of

- a. DSP
- b. DAC
- c. ADC
- d. SAR

Quiz

7. The circuit shown is a

- a. DSP
- b. DAC
- c. ADC
- d. SAR



Quiz

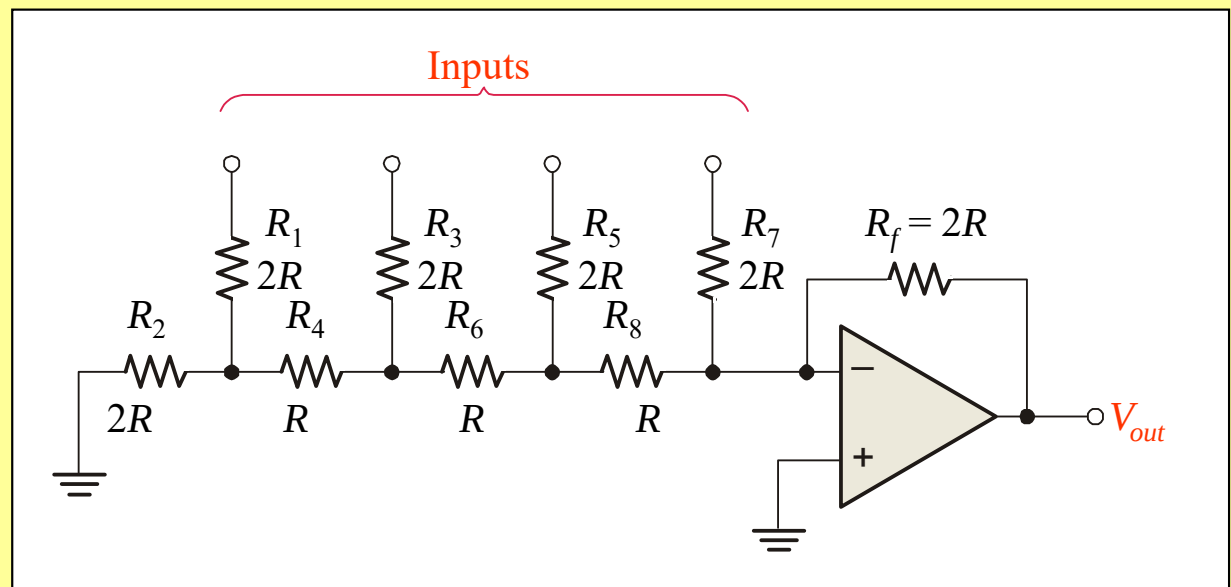
8. For the circuit shown, the input on the far left is for the

a. analog input

b. clock

c. LSB

d. MSB



Quiz

9. A reconstruction filter

- a. is a low-pass filter
- b. can have the same response as an anti-aliasing filter
- c. smoothes the output from a DAC
- d. all of the above

Quiz

10. A DSP is a specialized microprocessor that
- a. has a very large instruction set
 - b. is designed to be very fast
 - c. has internal anti-aliasing and reconstruction filters
 - d. all of the above